

- **Anterior communicating artery unites the two ACA.**

Lecture (8): Meninges and ventricular system I

Meninges

- The meninges are a series of membranes that surround the brain and spinal cord.
- They are made up of **three distinct layers**, called the dura mater, the arachnoid mater and the pia mater, each of which has a specific role.

Meninges of the Brain:

1- Pia Mater: (pia = gentle).

- Pia is the innermost layer of Meninges. It is a very delicate membrane.
- It is highly vascularized, with small blood vessels, and is permeable to CSF. Where small arteries enter the brain, the pia mater is carried down for short distances into the perivascular space.
- It invests the brain and extends to the depths of fissures and sulci.
- It forms the tela choroidea of the 3rd & 4th ventricles and fuses with the ependyma to form the choroid plexuses.
- It extends out over the cranial nerves and fuses with their epineurium.

2- Arachnoid Mater: (arachnoid = spider-like).

- It is a delicate membrane lying immediately under the dura mater, separated only by the narrow **subdural space**, which is filled by a film of fluid.
- It is separated from the pia mater by the **subarachnoid space**, which is filled with cerebrospinal fluid. Extending from the arachnoid mater across the subarachnoid space to the pia mater are numerous thin extensions, called **trabeculae, whose web-like appearance** gives the arachnoid mater its name.
- All the **cerebral arteries and cranial nerves** lie in **subarachnoid space**.
- Folds of the arachnoid mater project through the dura mater into the **superior dural sinus**. These are called **arachnoid villi or arachnoid granulations**. These are involved in the reabsorption of CSF into venous blood.

Subarachnoid cisternae:

- The arachnoid and pia are widely separated to form dilated areas called subarachnoid cisternae. All the cisternae are in free communications with the subarachnoid space:

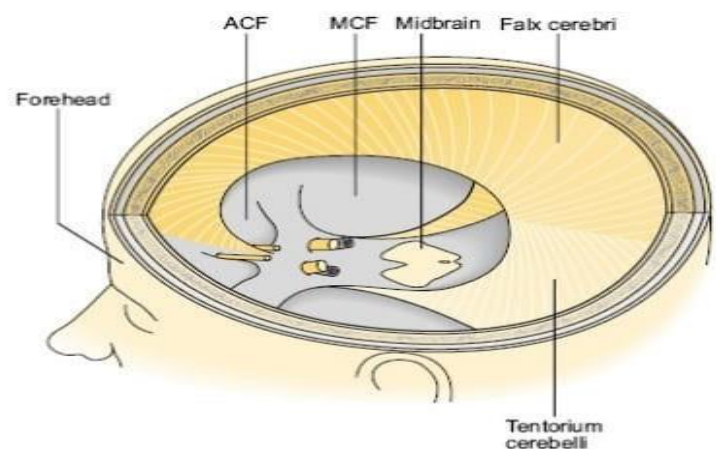
- a) **Cerebellomedullary cistern (cistern magna)**: is the largest, lies between the inferior surface of the cerebellum and the back of medulla. It contains **vertebral artery**.
- b) **Pontine cistern**: lies between the clivus and front of pons. It contains **basilar artery**.
- c) **Interpeduncular cistern**: lies between the two cerebral peduncles. It contains **circle of Willis**.

3- **Dura mater**. (dura = hard or tough).

- Dura is thick, leathery membrane forms the outer protective layer surrounding the brain. the **outermost layer** of meninges.
- It is formed from **two layers: the periosteal layer**, which lies close to the periosteum (the inner surface of the skull), and the **meningeal layer**, which lies closer to the brain.
- In most areas of the brain, these two layers are **fused**, but in some regions, they enclose blood-filled cavities called **dural sinuses**.

Dural Folds:

- The meningeal layer extends inward **prolongations** into several places to form partition the brain into separate areas and limit the movement of the brain within the cranial cavity.
- They divide the cranial cavity into freely communicating spaces, that lodge various parts of the brain.



Drawing of the meninges showing the falx cerebri and tentorium cerebelli with the vault of the cranium removed. MCF, middle cranial fossa; ACF, anterior cranial fossa.

1) **Falx cerebri**: (falx = sickle)

- The falx cerebri is a sickle-shaped sheet that extends into the longitudinal fissure, separating the two cerebral hemispheres.
- Its **apex** is attached to frontal crest & crista galli.
- Its **base** blends with the tentorium cerebelli and lodges **straight sinus**.
- Its **upper convex attached border** lodges **superior sagittal sinus**.
- Its **lower concave free border** curves over the corpus callosum.
- Its lodges **inferior sagittal sinus**.

2) Falx cerebelli:

- A small, **sickle-shaped fold** between two cerebellar hemispheres.
- It contains the **occipital sinus**.

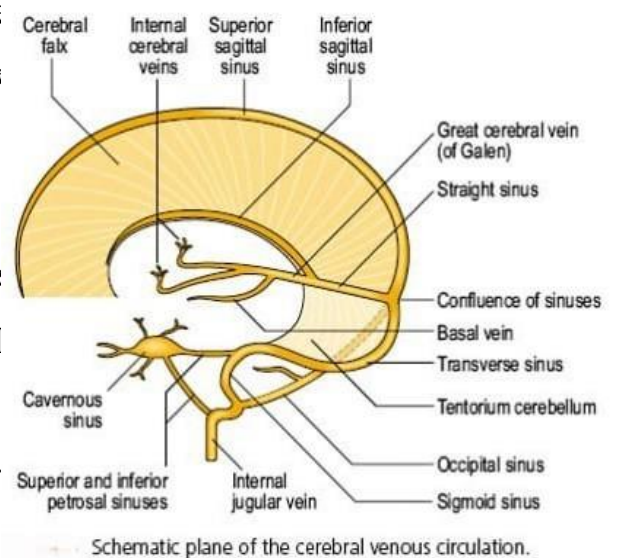
3) Tentorium cerebelli:

- It is a **tent-shaped fold** that extends horizontally separating the occipital lobes of cerebrum from upper surface of cerebellum, arching over posterior cranial fossa.
- Its convex **attached border** is attached to the posterior clinoid processes, superior petrosal sulcus and transverse sulcus.
- Its **free edge** forms a **U-shape**, enclosing the midbrain and attaching at the front to the anterior clinoid processes of the sphenoid bone.
- It contains **transverse sinuses & superior petrosal sinuses**.

4) Diaphragma sellae: is a small horizontal sheet separating the **pituitary gland**.

Dural sinuses:

- **All the dural sinuses**, except inferior sagittal and great cerebral vein, are formed by the union of inner and outer layers of the dura. Their walls are thin. They have no valves.
- **Emissary veins** are also valveless, connect the dural sinuses with the veins of the skull and with the veins of the scalp.



- Unpaired (single) sinuses:

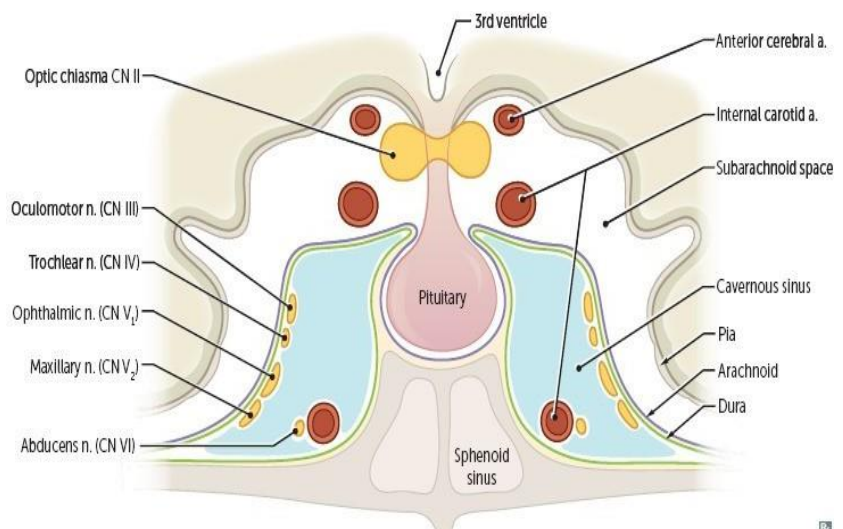
- **Superior sagittal sinus** in upper border of falx cerebri.
- **Inferior sagittal sinus** in lower border of falx cerebri.
- **Straight sinus** occupies the line of junction of falx cerebri with tentorium cerebelli.
- **Occipital sinus** in the posterior border of the falx cerebelli.

- Paired sinuses:

- Rt & Lt sphenoparietal sinuses.
 - Rt & Lt cavernous sinuses.
 - Rt & Lt superior petrosal sinuses.
 - Rt & Lt inferior petrosal sinuses.
 - Rt & Lt transverse sinuses.
 - Rt & Lt sigmoid sinuses.
- **R transverse sinus** is continuous with **superior sagittal sinus**, and the **left one** is continuous with **straight sinus**.
 - The sigmoid sinuses are a direct continuation of the transverse sinuses.
 - **Sigmoid sinus** becomes continuous with **internal jugular veins**.

Cavernous sinuses:

- They are **paired sinuses**, situated in the **middle cranial fossa** on each side of the body of the sphenoid bone.
- The sinus is **spindle-shaped** in lateral view. It is about 2 cm long and 1 cm wide. Numerous trabeculae cross their interior (**spongy appearance**).



➤ Relations:

- **Anterior:** superior orbital fissure.
- **Posterior:** apex of petrous part of temporal bone.
- **Lateral:** temporal lobe of brain. Structures embedded within the lateral wall (3rd, 4th CNs and ophthalmic and maxillary branches of 5th CN).
- **Medial:** pituitary gland, body of sphenoid & sphenoid air sinus.
- **Superior:** internal carotid artery.
- **Inferior:** Body of sphenoid & sphenoid air sinus.

➤ Contents:

Structures lying within the cavity (**internal carotid artery & abducent nerve**).

➤ Tributaries of cavernous sinus:

- Sphenoparietal sinus.
- Ophthalmic veins & central vein of retina.
- Superficial middle cerebral vein.
- Emissary veins from pterygoid venous plexus.
- Hypophyseal veins.
- Inter-cavernous sinuses

➤ Drainage of cavernous sinus:

1- Superior petrosal sinus

2- Inferior petrosal sinus.

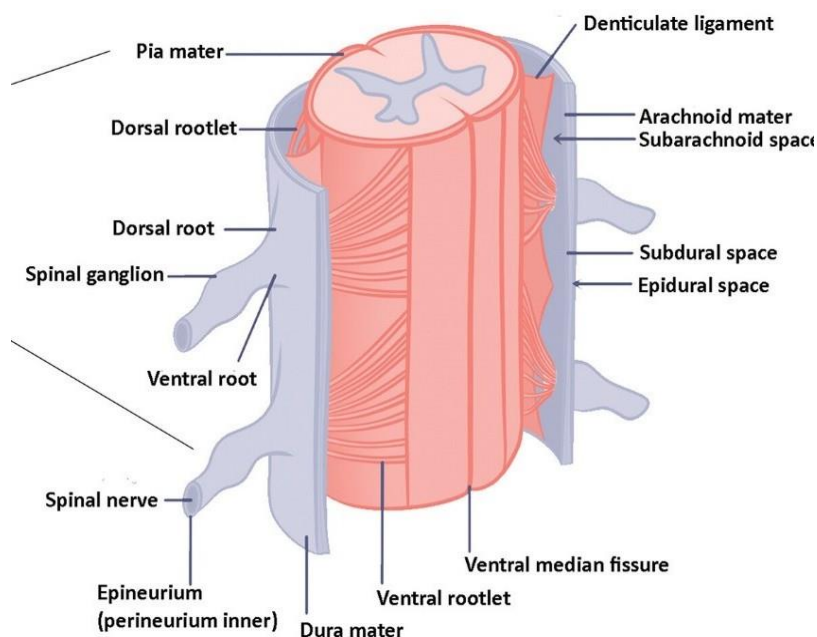
Spinal meninges:

• Spinal pia mater:

- It invests the surface of spinal cord and lines the anterior median sulcus.
- It is prolonged over the spinal nerve roots.
- It has **two** extensions:
 - 1) **Filum terminale** (extends from the conus medullaris to the back of the coccyx for fixation of spinal cord).
 - 2) **Denticulate ligaments** are triangular lateral extensions between the anterior and posterior roots of spinal nerves connecting to outer dura mater.

- Spinal arachnoid mater:

- lies between Pia mater internally and dura mater externally.
- It is separated from the pia mater by subarachnoid space, filled with cerebrospinal fluid.
- It is continuous above through the foramen magnum with the arachnoid covering the brain.
- Ends at the level of the **S2 vertebra**.



- Spinal dura mater:

- It is the outermost layer, enclosing the spinal cord and the cauda equina.
- The spinal dura is continuous above through the foramen magnum with meningeal layer of dura covering the brain.
- Spinal dura ends inferiorly at level of the **S2 vertebra**.

- Lumbar cistern:

- Below the conus medullaris, down to **S2 vertebra**.
- It lodges the **nerves of the cauda equina**.
- It is a site for lumbar puncture.

